UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 008 · Problems on Taylor's and Maclaurin's series · compiled 14 October 2026

Contents

1. Short notes
2. Keyword index
3. Rapid recall
4. Last 10 minutes

1 Short notes

1 · Taylor expansion about a point (a, b)

About (a,b): $h=x-a$, $k=y-b$; table of $f, f_x, f_y, f_{xx}, f_{xy}, f_{yy}$ at the point; $f+\{hf_x+kf_y\}+\frac{1}{2!}\{h^2f_{xx}+2hkf_{xy}+k^2f_{yy}\}$. $e^x\cos y$ at $(1,\frac{\pi}{4})$: $\frac{e}{\sqrt{2}}[1+h-k+\frac{1}{2}(h^2-2hk-k^2)]$. $\sin(xy)$ at $(1,\frac{\pi}{2})$: $1-\frac{\pi^2}{8}h^2-\frac{\pi}{2}hk-\frac{1}{2}k^2$. $xy^2+y\cos(x-y)$ at $(1,1)$: $2+h+3k+\frac{1}{2}(-h^2+6hk+k^2)$.

2 · Maclaurin expansion about the origin

About $(0,0)$: $f+\{xf_x+yf_y\}+\frac{1}{2!}\{\dots\}+\frac{1}{3!}\{\dots\}$. $\log(2x+y+1)=(2x+y)-\frac{(2x+y)^2}{2}$; $e^{x-y}=1+(x-y)+\frac{(x-y)^2}{2}+\frac{(x-y)^3}{6}$; $e^x\log(1+y)=y+xy-\frac{y^2}{2}+\frac{x^2y}{2}-\frac{xy^2}{2}+\frac{y^3}{3}$. Check one-combination functions with the one-variable series.

3 · Approximate values from a Taylor polynomial

Expand about a nearby easy point and substitute small h, k with signs. $\tan^{-1}(xy)$ at $(0.9,-1.2)$ from $(1,-1)$: -0.835 (first order), -0.823 (second); exact -0.824. $\tan^{-1}\frac{0.9}{1.1}\approx 0.6854$.

2 Keyword index

14 terms; the number in brackets is the topic.

Approximate value (3). Truncated series at a nearby point.

Common factor (1). $\frac{e}{\sqrt{2}}$ in $e^x \cos y$ at $(1, \frac{\pi}{4})$.

Expansion point choice (3). Near the target, with exact values.

In powers of (x − a), (y − b) (1). About the point (a, b).

Increment signs (3). $h = 0.9 - 1 = -0.1$.

Maclaurin's series (2). About the origin.

One-combination check (2). $f = g(px + qy)$: use g's series.

Order of accuracy (3). More terms, smaller error.

Product of series (2). $e^x \log(1 + y)$, $e^{ax} \sin by$.

Taylor's series about (a, b) (1). E1.1.

Third-order partials (1). Needed for third-degree terms.

Up to n-th degree (1). Keep all terms of total degree $\leq n$.

Up to three terms (2). Three non-zero terms.

Vanishing derivatives (1). $\sin(xy)$ at $(1, \frac{\pi}{2})$ has $f_x = f_y = 0$.

3 Rapid recall

1	"In powers of $(x-1)$ and $(y-\frac{\pi}{4})$" means?	About $(1, \frac{\pi}{4})$.
2	$e^x \cos y$ derivatives at $(1, \frac{\pi}{4})$?	All $\pm\frac{e}{\sqrt{2}}$.
3	$e^x \cos y$ to second degree?	$\frac{e}{\sqrt{2}}[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)]$.
4	$\log(2x + y + 1)$, three terms?	$2x + y - 2x^2 - 2xy - \frac{y^2}{2}$.
5	"Up to three terms" means?	Three non-zero terms.
6	$\tan^{-1}(xy)$ about $(1, -1)$?	$-\frac{\pi}{4} - \frac{1}{2}h + \frac{1}{2}k + \frac{1}{4}(h^2 + k^2)$.
7	$\tan^{-1}(0.9 \times (-1.2))$ to first order?	$-\frac{\pi}{4} - 0.05 = -0.835$.
8	... to second order?	-0.823 (exact -0.824).
9	e^{x-y} about the origin?	$1 + (x-y) + \frac{(x-y)^2}{2} + \frac{(x-y)^3}{6}$.
10	$f_{xy}(0,0)$ for $e^x \log(1+y)$?	1 (the slide prints 0).
11	First six terms of $e^x \log(1+y)$?	$y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$.
12	$\sin(xy)$ about $(1, \frac{\pi}{2})$?	$1 - \frac{\pi^2}{8}h^2 - \frac{\pi}{2}hk - \frac{1}{2}k^2$.
13	$f_{yy}(1,1)$ for $xy^2 + y\cos(x-y)$?	1 (the slide prints 3).
14	$xy^2 + y\cos(x-y)$ about $(1,1)$?	$2 + h + 3k + \frac{1}{2}(-h^2 + 6hk + k^2)$.
15	$\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$, third degree?	$\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2 - \frac{1}{12}(h+k)^3$.
16	$\log(1 + x - y)$ to third degree?	$(x-y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3}$.
17	$e^{ax}\sin by$ to third degree?	$by + abxy + \frac{a^2 b}{2}x^2 y - \frac{b^3}{6}y^3$.
18	$\sin(x + 2y)$ to third degree?	$(x+2y) - \frac{(x+2y)^3}{6}$.
19	$\tan^{-1}\frac{0.9}{1.1}$ by Taylor?	≈ 0.6854.

Mnemonic (study aid). *"Series, table, values, substitute, check."* The five steps of every class 008 problem (Figure F1.1).

4 Last 10 minutes before the exam

Top facts

1. Method: general series → derivative table → values at the point → substitute → check.
2. $h = x - a$, $k = y - b$; second order $\frac{1}{2!}\{h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}\}$; third order $\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$.
3. $e^x \cos y$ at $(1, \frac{\pi}{4})$: $\frac{e}{\sqrt{2}}[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)]$ (ESA May 2024).
4. Functions of one combination $(2x + y,\ x - y,\ x + 2y)$: use the one-variable series.
5. Slide errors: $f_{yy}(1,1) = 1$ (not 3) for $xy^2 + y\cos(x - y)$; $f_{xy}(0,0) = 1$ (not 0) for $e^x \log(1 + y)$.
6. Approximation: expand about a nearby easy point; keep second order (-0.823 vs exact -0.824).
7. ESA: $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$ (2nd and 3rd degree), $e^x \cos y$, $\log(1 + x - y)$, $e^{ax} \sin by$, $\sin(x + 2y)$.

Top formulas

E1.1 · Taylor's series about (a, b), to second order ESA ★★★

$$f(x,y) = f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}\} + \ldots, \quad h = x - a,\ k = y - b$$

GIVES	The polynomial to substitute into.
APPLIES	All derivatives evaluated at (a, b).
NOTE	h, k are shorthand added for clarity.

S008·3 · S008·14 · T008 00:00:37 · T008 00:01:25

E1.2 · Third-degree term ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

GIVES	The terms for "up to third degree".
APPLIES	Add to E1.1 or E2.1.
NOTE	Coefficients 1, 3, 3, 1; asked in four of six ESA questions.

S008·10 · ESA Dec-23 1b · ESA Jun-24 1c

E2.1 · Maclaurin's series (about the origin) ESA ★★★

$$f(x,y) = f(0,0) + \{xf_x + yf_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}\} + \ldots$$

GIVES	The polynomial about $(0, 0)$.
APPLIES	Derivatives at $(0, 0)$.
NOTE	Slide 008·10 calls this "Taylor's series about (0,0)"; it is the same thing.

S008·10 · S008·12 · T008 00:06:58

E2.2 · One-combination check ESA ★★

$$f = g(t),\ t = px + qy \implies f(x,y) = g(0) + g'(0)\,t + \frac{g''(0)}{2!}t^2 + \frac{g'''(0)}{3!}t^3 + \dots$$

GIVES The same expansion in one line.

APPLIES f depends on x and y only through $px + qy$.

NOTE $\log(2x + y + 1) = t - \frac{t^2}{2} + \frac{t^3}{3}$ with $t = 2x + y$; $e^{x-y} = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}$ with $t = x - y$.

ADDED FOR CLARITY S008·6 · S008·11

E3.1 · Approximation by a Taylor polynomial ESA ★★

$$f(a+h, b+k) \approx f(a,b) + hf_x + kf_y + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\}$$

GIVES A numerical estimate.

APPLIES h, k small; (a,b) a point where f and its derivatives are easy.

NOTE Choose (a,b) with "nice" values: $(1,-1)$ for $(0.9,-1.2)$.

S008·7 · S008·9 · T008 00:12:10